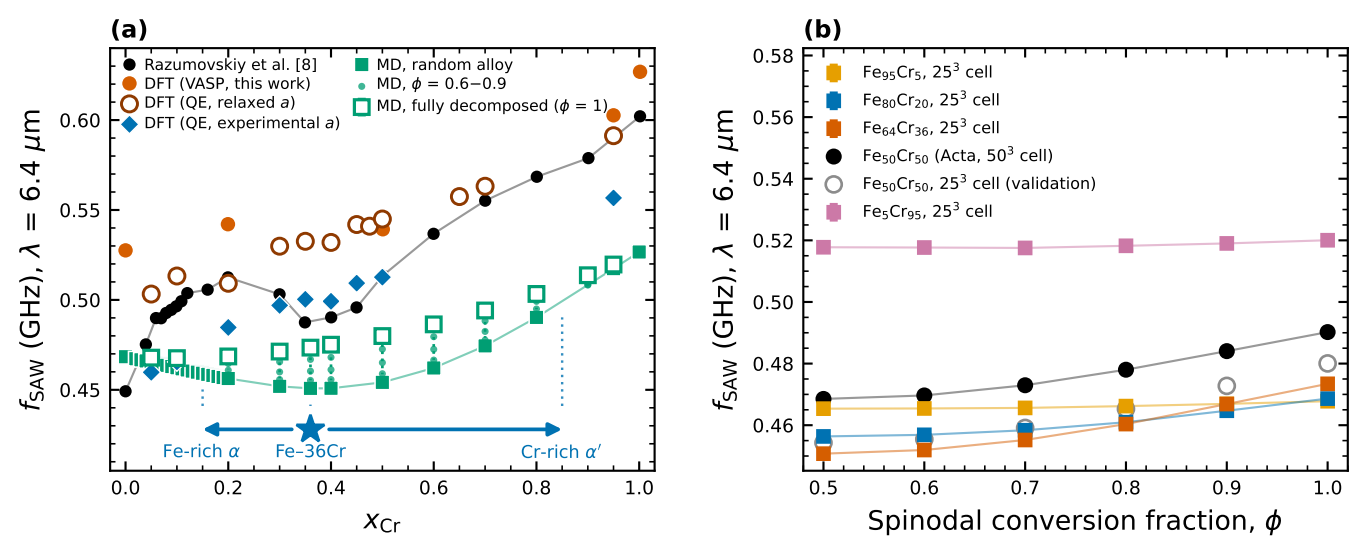
(a)
Razumovskiy et al. [8]
DFT (VASP, this work)
DFT (QE, relaxed a)
DFT (QE, experimental a)
MD, random alloy
MD, ϕ = 0.6−0.9
MD, fully decomposed (ϕ = 1)
Fe-rich α
Fe–36Cr
Cr-rich α′
f_SAW (GHz), λ = 6.4 μm
x_Cr
0.45
0.50
0.55
0.60
0.0
0.2
0.4
0.6
0.8
1.0
(b)
Fe95Cr5, 25³ cell
Fe80Cr20, 25³ cell
Fe64Cr36, 25³ cell
Fe50Cr50 (Acta, 50³ cell)
Fe50Cr50, 25³ cell (validation)
Fe5Cr95, 25³ cell
f_SAW (GHz), λ = 6.4 μm
Spinodal conversion fraction, ϕ
0.46
0.48
0.50
0.52
0.54
0.56
0.58
0.5
0.6
0.7
0.8
0.9
1.0